Ch 4

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★ Memory Management

- → To alieve a degree of multiprogramming & proper utilization of memory, memory management is important
- → Handles or manages primary memory & moves processes back & forth btw main memory (RAM) & disk during execution
- → keeps track of each & every memory location (free or allocated)
- → checks how much memory is to be allocated to processes.
- → tracks freed & unallocated memory

OS takes care of mapping the logical addresses to physical addresses at time of memory allocation - Address Binding

- Symbolic Add. (Address used in source code; variable names, constants, etc.)
- Relative Add. (at time of compilation, compiler converts symbolic address → relative address)
- Physical Add. (Loader generates these addresses when program is loaded into the memory)

★ Logical & Physical Address

S.No.	Logical Address	Physical Address
1.	Generated by the CPU	A location in the memory unit
2.	Address space consists of all sets of logical address.	Address space consists of all Physical address mapped to logical addresses
3.	Generated by CPU for with reference to a specific program	computed using MMU (Memory Management Unit)
4.	User has the ability to view	User can't view the Physical Address of a program directly.

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5.	User can use logical address to access the Physical Address.	User can indirectly access the Logical address

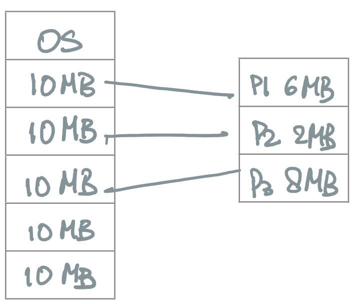
Similarities -

- Both are used to identify a specific location in memory
- Both can be represented in different formats like Binary, Hexadecimal or Decimal
- Both have finite range (Determined by no. of bits used to represent them)

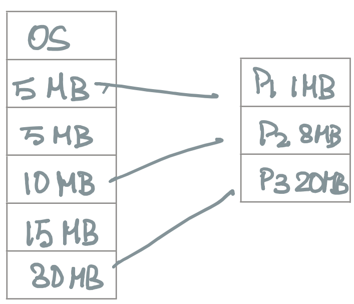
★ Contiguous Memory Allocation

- A method in which a single contiguous section/part of memory is allocated to a process or file needing it.
  (resides in same place & not distributed in random fashion)

Static / Fixed Partition



Dynamic / Variable Partition



→ Advantages : Faster Execution (less time consumed)
Minimal Overhead (fewer address translations)
Easier to Control (sequential)

→ Disadvantages : Internal & External Fragmentation
Limited Multiprogramming
Wastage of Memory

★ Non Contiguous Memory Allocation

- Allocates memory space present in different locations to the process as per its

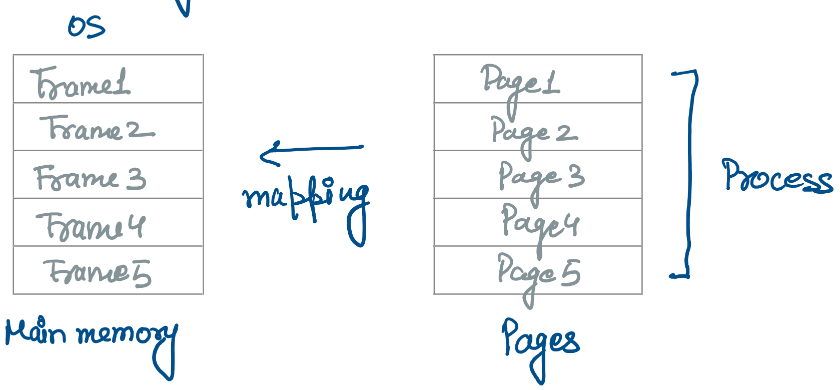
- Allocates memory space present in different locations to the process as per its requirements

-> Advantages : No memory wastage
Increased Multiprogramming
Reduced Fragmentation (minimizes internal fragmentation)

-> Disadvantages : Slower Execution

★ Paging

- A storage mechanism used in OS to retrieve processess from secondary storage to the main memory as pages.
- Primary concept ; break each process into pages. Primary memory would also be separated into frames.
- fixed size partitioning scheme



★ Memory Management Unit (MMU)

MMU is responsible for converting logical Address into Physical Addresses.

- When CPU accesses a page using its logical address, the OS must first collect the physical address in order to access that page physically.

Logical Address <
- Page no. - no. of bits used to represent
- offset (page size in logical address)

★ TLB (Transation Look Aside Buffer)

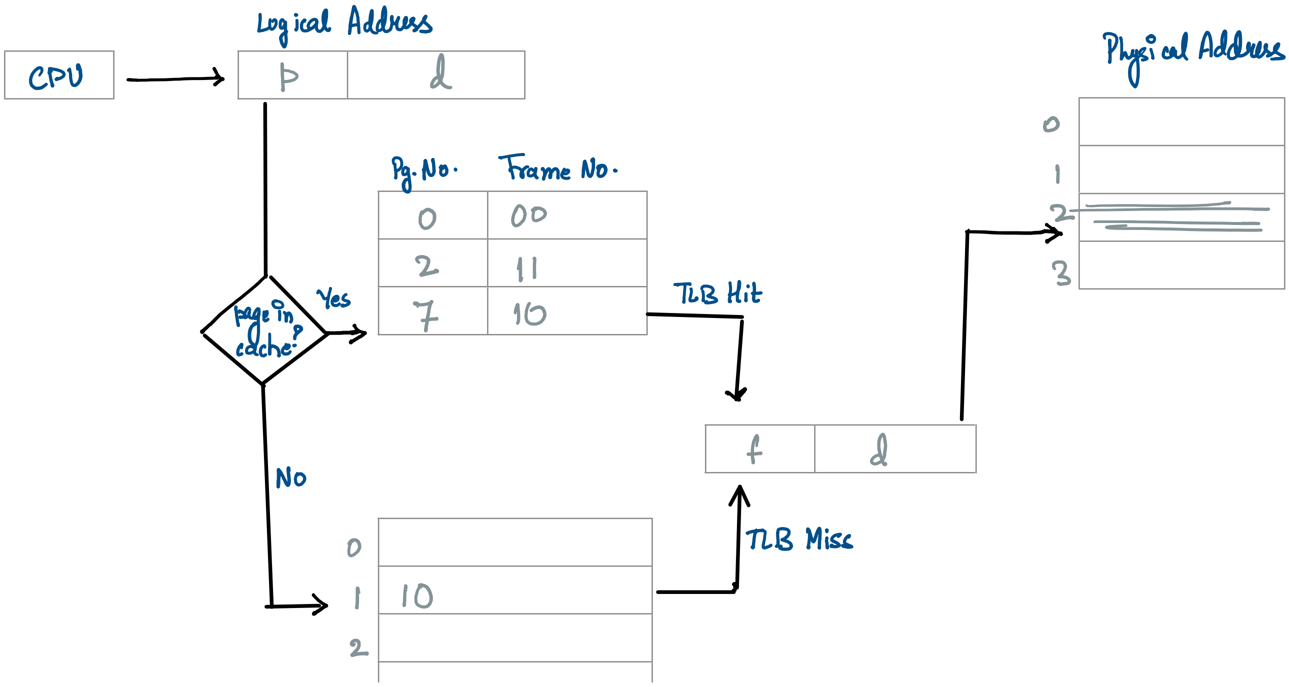
- special cache used to keep track of recently used transactions
- contains page table entries that have been most recently used
- page table entry found (TLB Hit)
- page table entry not found (TLB Miss)
- TLB checks if the page is already in Main Memory; if not then a page fault is issued & TLB is updated to include the new page entry.

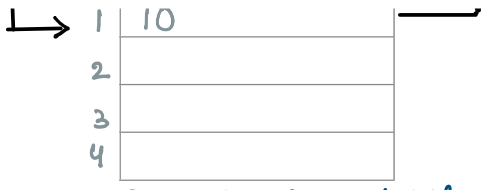
→ TLB Hit

i) CPU generates a virtual (logical address)
ii) checked in TLB (present)
iii) Corresponding frame no. is retrieved, which now tells where the main memory page lies.

→ TLB Miss

(i) CPU generates a virtual (logical address)
(ii) Checked in TLB (not present)
(iii) Now the page no. is matched to the page table residing in the main memory
(iv) Corresponding frame no. is retrieved
(iv) TLB updated with new PTE





Page Table or Page Map Table

★ Inverted Page Table (IPT)

IPT is a datastructure used to map physical memory to virtual memory pages

Page Table			Inverted Page Table		
Virtual Address (Pages)	→	Physical Address (Frames)	Physical Address (Frames)	→	Virtual / Logical Address (Pages)

★ Segmentation

- segmentation is dividing into segments. (Process → segments) not necessarily of same sizes.

Virtual Memory Segmentation

each process → segments; not done all at once

May or may not take place at run time

Simple Segmentation

Process → segments; all loaded into main memory at run time

not necessarily contiguously

→ Segment Table

	Base Add.	Limit
0	500	600
1	2500	800
2	1500	200
3	3800	400

↙ Segment number

Segment Table

Physical Addr. Space

Virtual Memory
(managed by MMU)

Paging

Segmentation

✱ Page Replacement Algorithms

- Techniques used in OS to manage memory efficiently when the virtual memory is full.
- These algorithms determine which existing page to replace.
- Common Page Replacement Algorithms -
  (i) FIFO
  (ii) Least Recently Used (LRU)
  (iii) Most Recently Used (MRU)
  (iv) Optimal Page Replacement